

Random walks

- **History of random walks**
- **Basic theory of random walks**
- **Einstein's paper: using random walks to calculate the size of a molecule**
- **Stock prices and random walks**

Random walk presentation papers posted tonight

Karl Pearson (1857-1936)

- **English mathematician**
- **Educated at King's College, Cambridge**
- **Founded the first university statistics department at University College London in 1911**



Karl Pearson in 1890

First use of the term “random walk”

The Problem of the Random Walk.

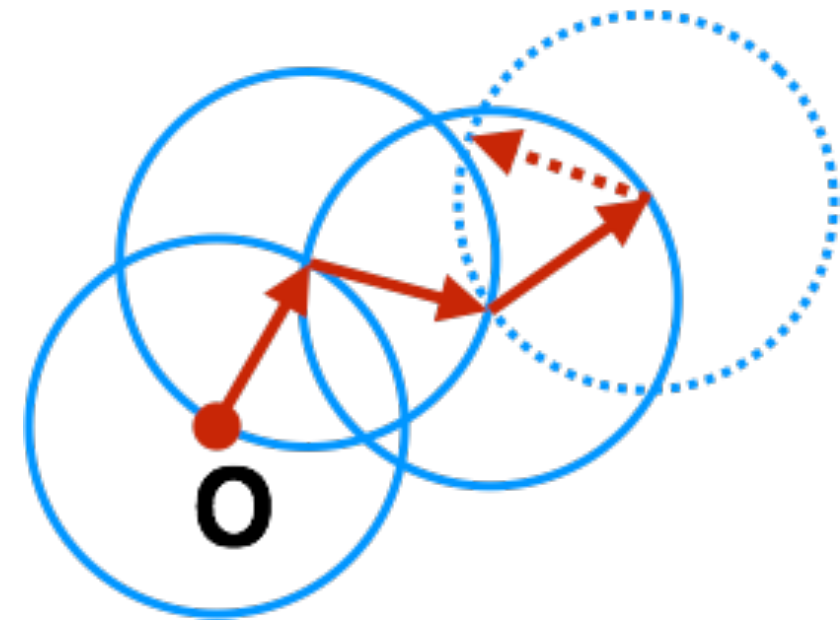
CAN any of your readers refer me to a work wherein I should find a solution of the following problem, or failing the knowledge of any existing solution provide me with an original one? I should be extremely grateful for aid in the matter.

A man starts from a point O and walks l yards in a straight line; he then turns through any angle whatever and walks another l yards in a second straight line. He repeats this process n times. I require the probability that after these n stretches he is at a distance between r and $r + \delta r$ from his starting point, O .

The problem is one of considerable interest, but I have only succeeded in obtaining an integrated solution for *two* stretches. I think, however, that a solution ought to be found, if only in the form of a series in powers of $1/n$, when n is large.

KARL PEARSON.

The Gables, East Ilsley, Berks.

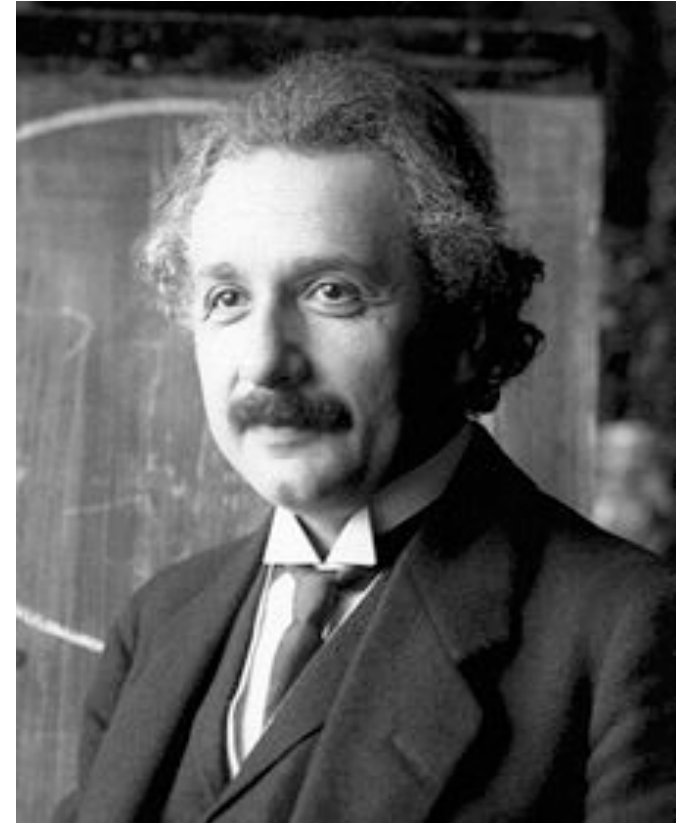


Einstein's paper

ON THE MOVEMENT OF SMALL PARTICLES SUSPENDED IN STATIONARY
LIQUIDS REQUIRED BY THE MOLECULAR-KINETIC THEORY OF HEAT

by A. Einstein

[*Annalen der Physik* 17 (1905): 549-560]



If Pearson had been a careful reader of the literature, he could have found the answer from the (not yet famous) Mr. Einstein!



Albert Einstein

Institute of Advanced Studies, Princeton

Physics

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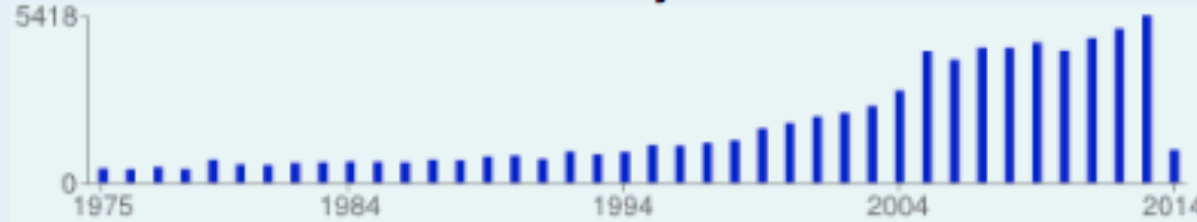
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Citations to my articles



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Title / Author	Cited by	Year
Can quantum-mechanical description of physical reality be considered complete? A Einstein, B Podolsky, N Rosen Physical review 47 (10), 777	12209	1935
Über einen die Erzeugung und Verwandlung des Lichtes betreffenden heuristischen Gesichtspunkt A Einstein Ann. Phys. 17, 132-148	6648 *	1905
On the movement of small particles suspended in stationary liquids required by the molecular-kinetic theory of heat A Einstein Annalen der Physik 17, 549-560	5330 *	1905
Zur Elektrodynamik bewegter Körper A Einstein	3613 *	
Investigations on the Theory of the Brownian Movement A Einstein Dover publications	3105	1956
The meaning of relativity A Einstein Routledge	2827	2003
Eine neue bestimmung der moleküldimensionen A Einstein Annalen der Physik 324 (2), 289-306	2771	1906

Despite being world-famous for developing special and general relativity, this is one of Einstein's most highly cited papers.

Actually, history is even more interesting

THÉORIE DE LA SPÉCULATION,

PAR M. L. BACHELIER.

INTRODUCTION.

Les influences qui déterminent les mouvements de la Bourse sont innombrables, des événements passés, actuels ou même escomptables, ne présentant souvent aucun rapport apparent avec ses variations, se répercutent sur son cours.

A côté des causes en quelque sorte naturelles des variations, interviennent aussi des causes factices : la Bourse agit sur elle-même et le mouvement actuel est fonction, non seulement des mouvements antérieurs, mais aussi de la position de place.

La détermination de ces mouvements se subordonne à un nombre infini de facteurs : il est dès lors impossible d'en espérer la prévision mathématique. Les opinions contradictoires relatives à ces variations se partagent si bien qu'au même instant les acheteurs croient à la hausse et les vendeurs à la baisse.

Le Calcul des probabilités ne pourra sans doute jamais s'appliquer aux mouvements de la cote et la dynamique de la Bourse ne sera jamais une science exacte.

L. BACHELIER

Théorie de la spéculation

Annales scientifiques de l'É.N.S. 3^e série, tome 17 (1900), p. 21-86.

http://www.numdam.org/item?id=ASENS_1900_3_17__21_0



Bachelier beat Einstein to the theory of random walks by 5 years – when working on Stock prices

Different problem, same result

L'expression définitive de la probabilité est donc

$$p = \frac{1}{\sqrt{2\pi k^2 t}} e^{-\frac{x^2}{2k^2 t}},$$

L'espérance mathématique

$$\int_0^\infty p x dx = k \sqrt{t}$$

est proportionnelle à la racine carrée du temps.

Bachelier's result

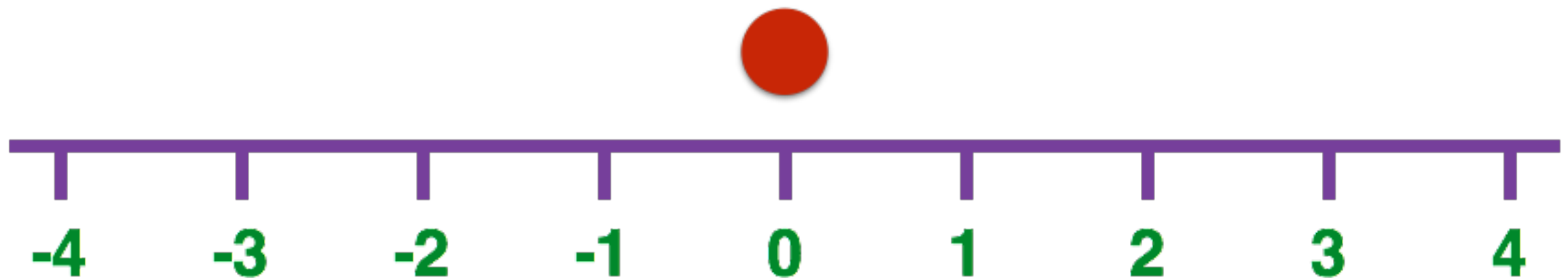
$$f(x, t) = \frac{n}{\sqrt{4\pi D}} \frac{e^{-\frac{x^2}{4Dt}}}{\sqrt{t}}.$$

$$\lambda_x = \sqrt{2Dt}.$$

Einstein's result

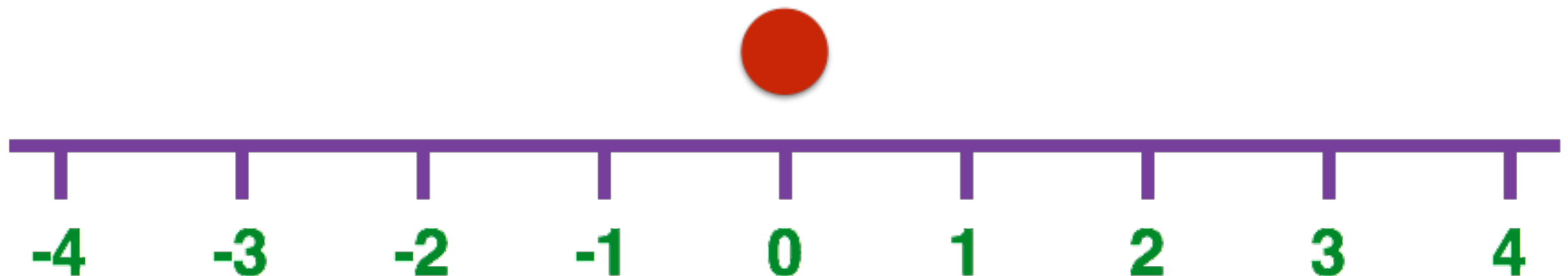
What is a random walk?

- Start at the origin



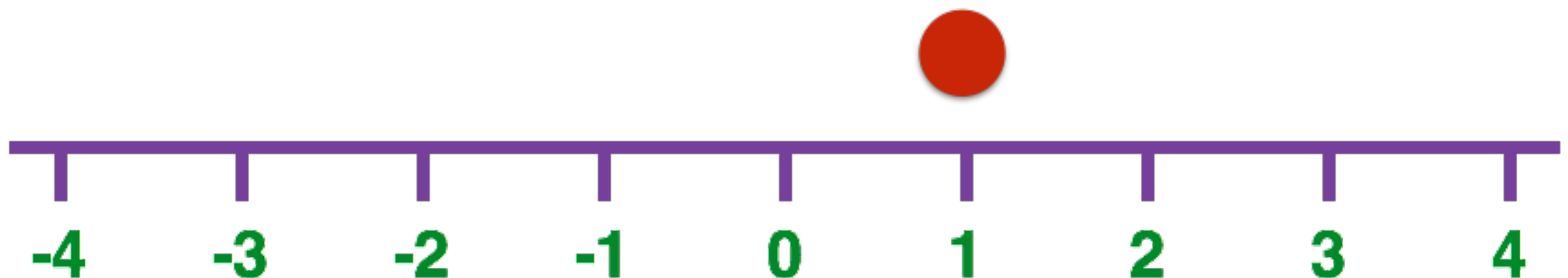
What is a random walk?

- Start at the origin
- Flip a fair coin



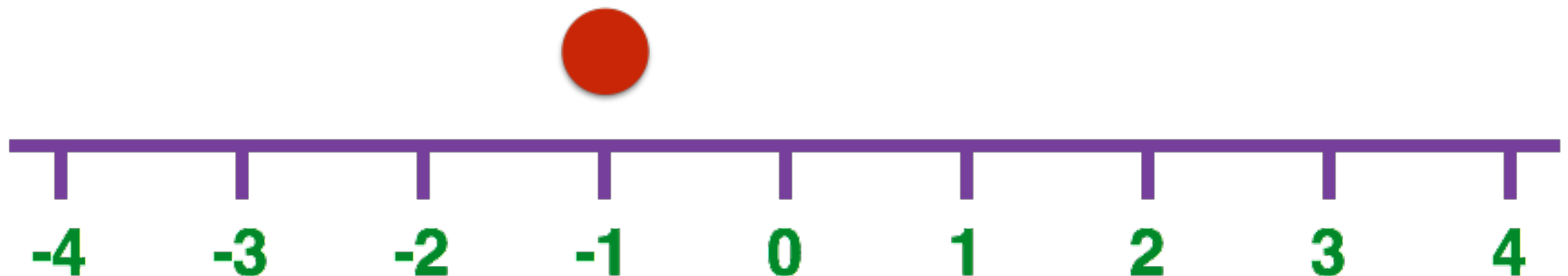
What is a random walk?

- **Start at the origin**
- **Flip a fair coin**
- **Walking rule: go right if heads**



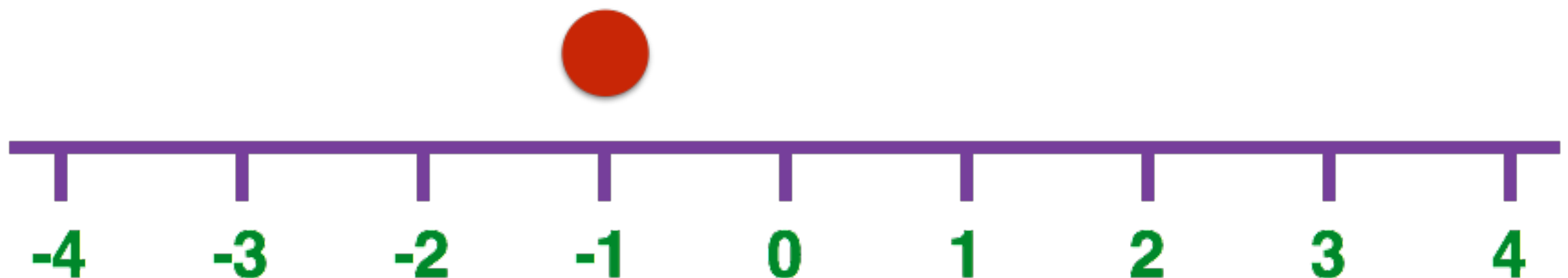
What is a random walk?

- Start at the origin
- Flip a fair coin
- Walking rule: go right if heads, left if tails



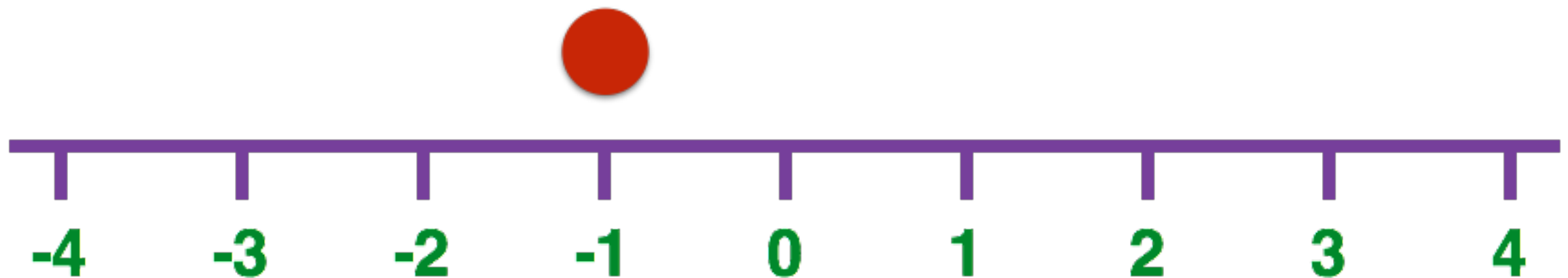
What is a random walk?

- **Start at the origin**
- **Flip a fair coin**
- **Walking rule: go right if heads, left if tails**
- **Prior history has no influence on next move**



Question

How far do you travel after time T ?



Random walk in Matlab

Input program

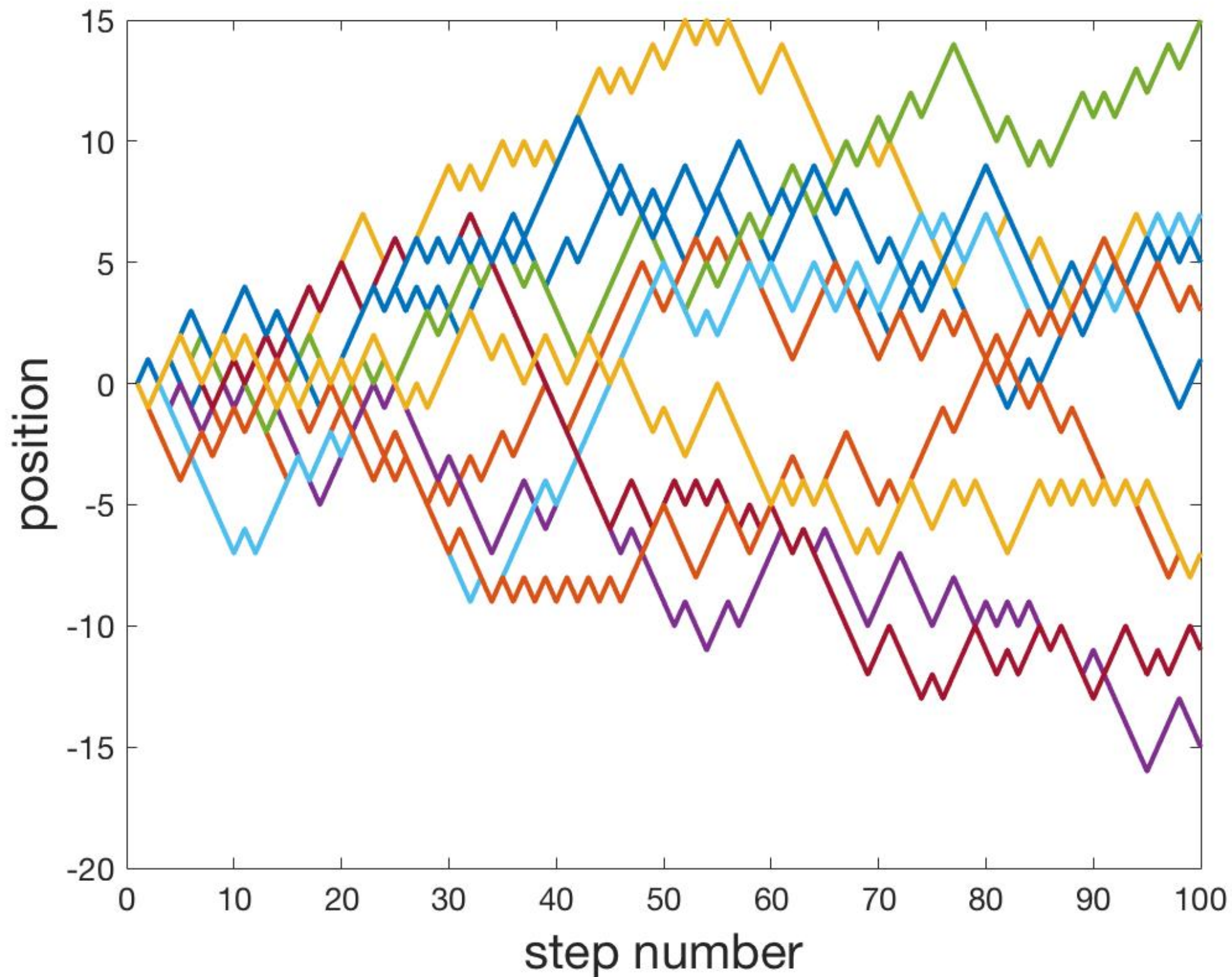
```
clear all;  
close all;  
  
%starting position  
x = 0;  
  
%number of steps  
steps = 10;  
  
%carry out steps  
for i = 1:steps  
    %print results  
    disp([i, x])  
    if rand(1,1) < .5  
        x = x+1;  
    else  
        x = x-1;  
    end  
end
```

randomwalk.m

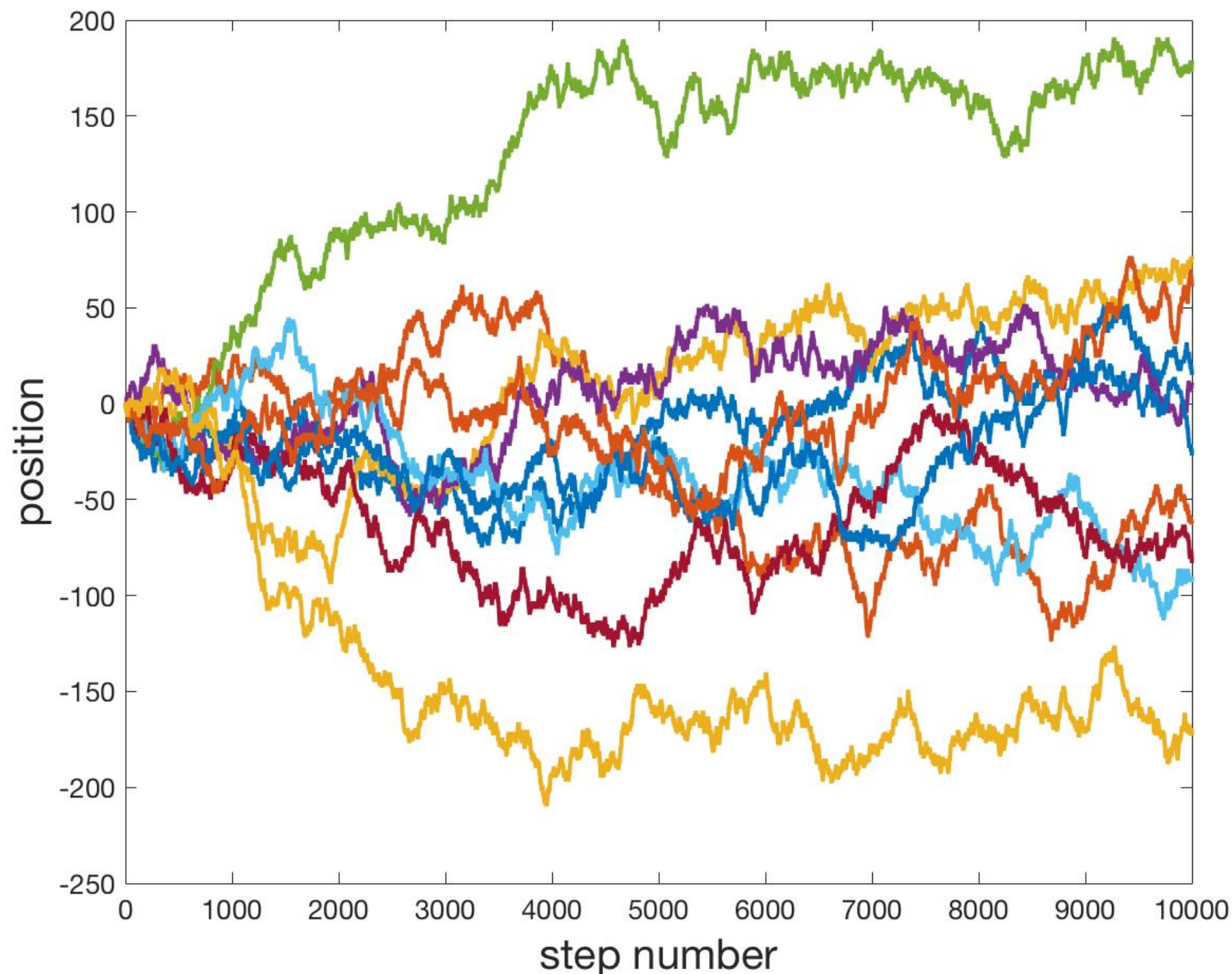
Output

```
>> randomwalk  
1      0  
  
2      1  
  
3      0  
  
4     -1  
  
5     -2  
  
6     -1  
  
7      0  
  
8     -1  
  
9     -2  
  
10     -1
```

Ten 100-step random walks

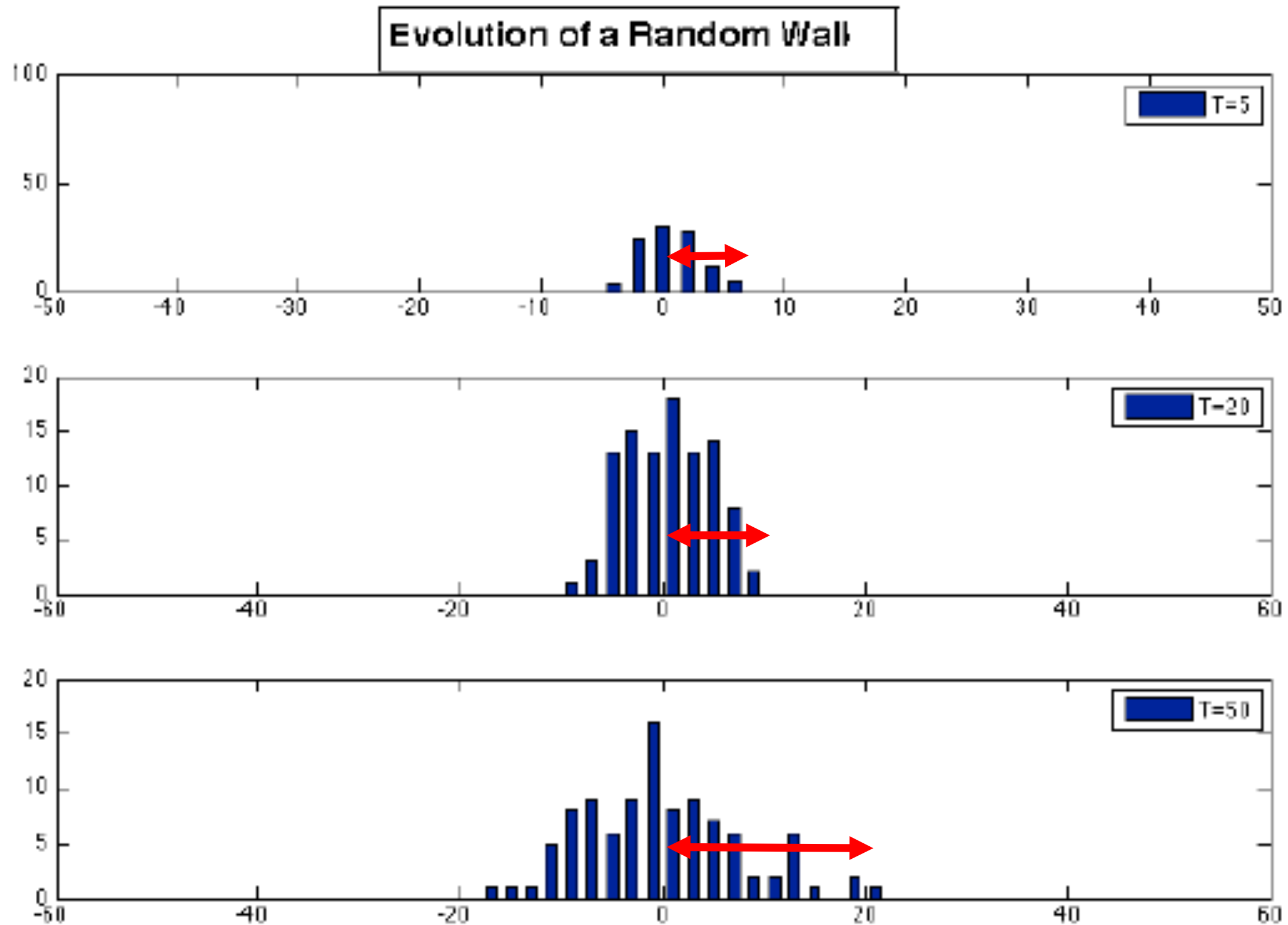


Ten 10000-step random walks

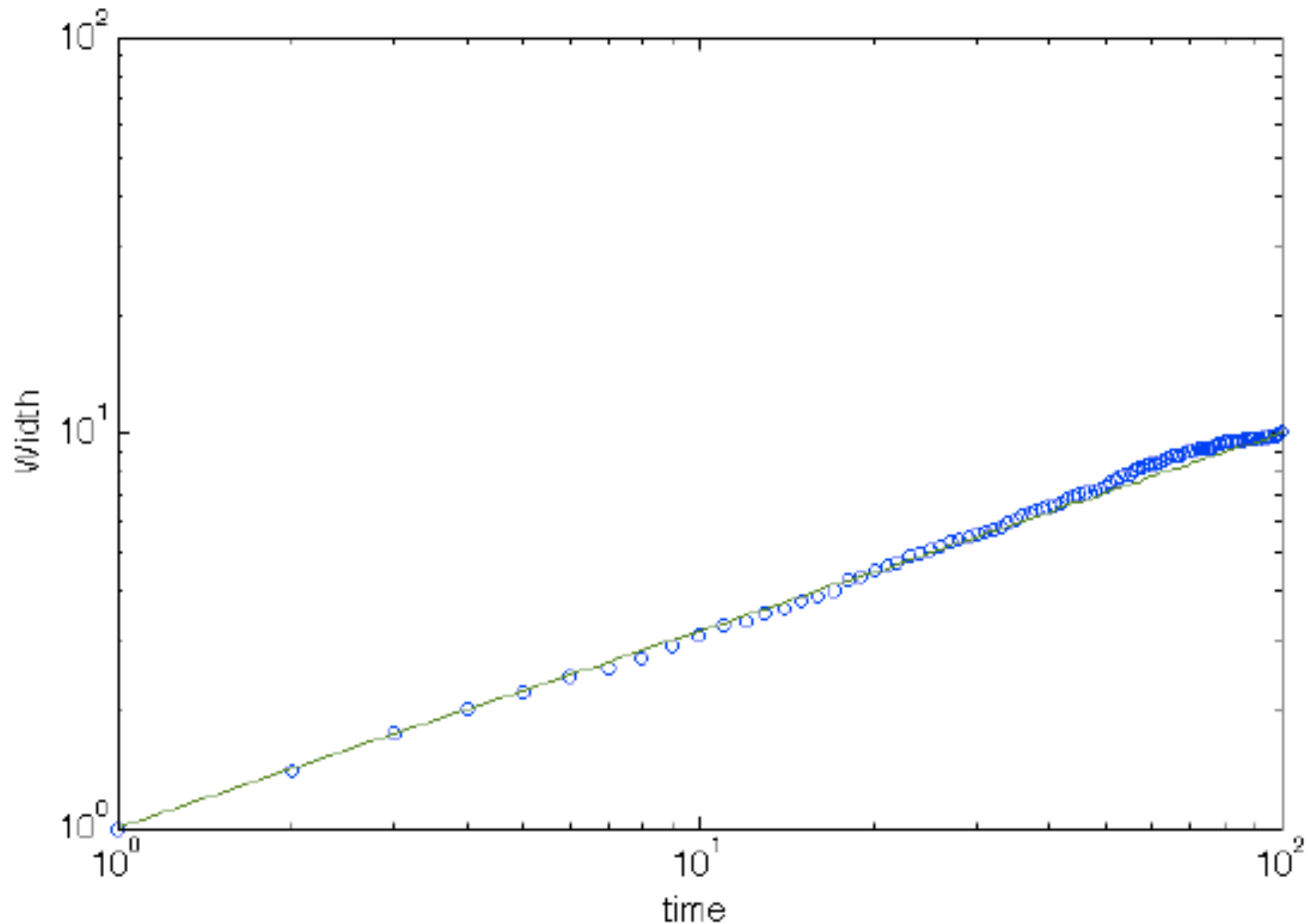


Matlab movie

Snapshots of the distribution



Width of the distribution

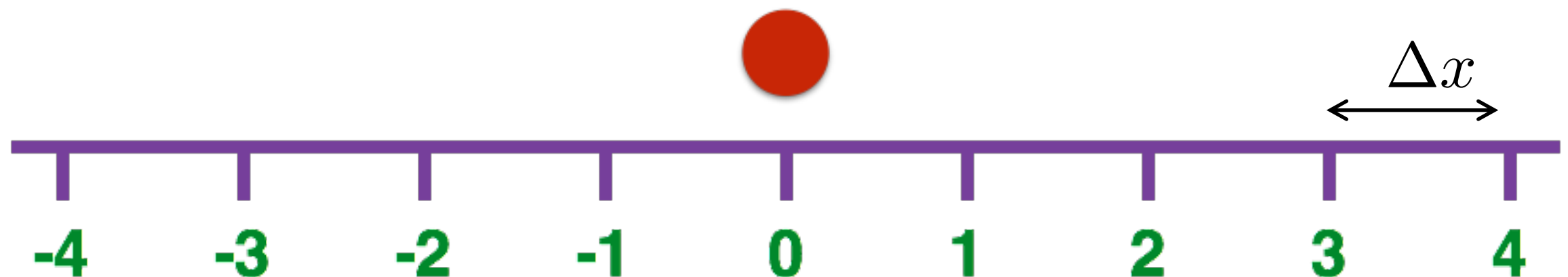


Turning walking rule into math

Walking rule: flip a fair coin and move right if heads, left if tails

Let p_i^n be the probability of being at position $i \Delta x$ at time $n \Delta t$

$$p_i^{n+1} = \frac{1}{2} (p_{i-1}^n + p_{i+1}^n)$$



The fundamental equation

$$p(x, t + \Delta t) = \frac{1}{2} (p(x - \Delta x, t) + p(x + \Delta x, t))$$

MATH!
(AM 105)



$$\frac{\partial p}{\partial t} = D \frac{\partial^2 p}{\partial x^2}$$



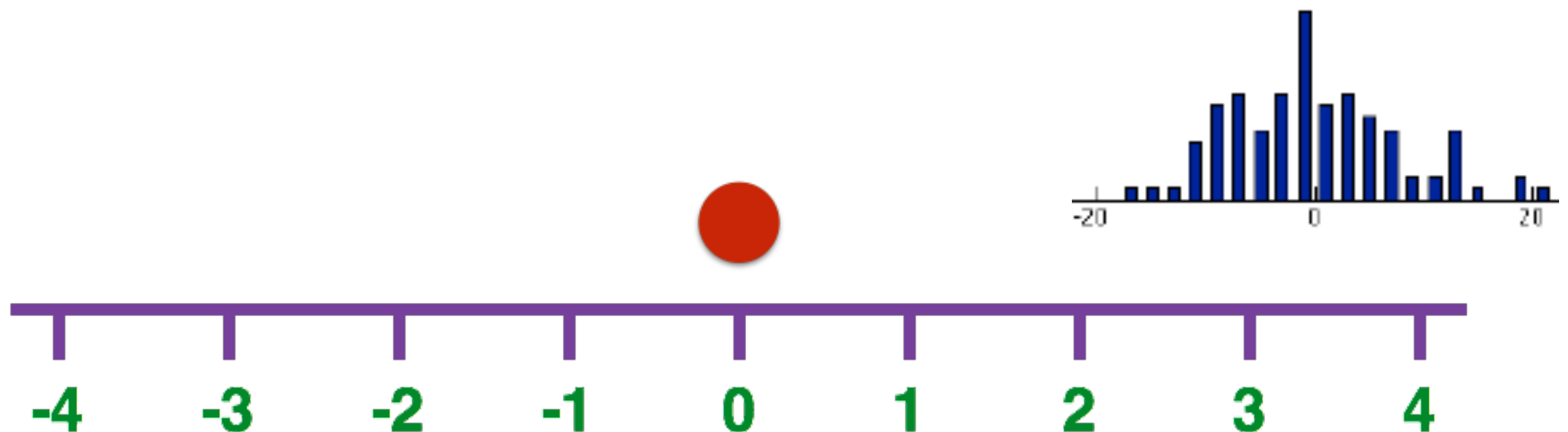
Diffusion/heat equation

$$\frac{\partial p}{\partial t} = D \frac{\partial^2 p}{\partial x^2}$$

$$p(x, t) = \frac{1}{\sqrt{4\pi Dt}} \exp\left(-\frac{x^2}{4Dt}\right)$$

$$D = \frac{1}{2} \frac{(\Delta x)^2}{\Delta t}$$

$$L = \sqrt{Dt}$$



Einstein's paper

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by A. Einstein

[*Annalen der Physik* 17 (1905): 549-560]

It will be shown in this paper that, according to the molecular-kinetic theory of heat, bodies of microscopically visible size suspended in liquids must, as a result of thermal molecular motions, perform motions of such magnitude that these motions can easily be detected by a microscope. It is possible that the motions to be discussed here are identical with the so-called "Brownian molecular motion"; however, the data available to me on the latter are so imprecise that I could not form a definite opinion on this matter.

If it is really possible to observe the motion to be discussed here, along with the laws it is expected to obey, then classical thermodynamics can no longer be viewed as strictly valid even for microscopically distinguishable spaces, and an exact determination of the real size of atoms becomes possible. Conversely, if the prediction of this motion were to be proved wrong, this fact would provide a weighty argument against the molecular-kinetic conception of heat.

Main points

1. **“Visible” (ie, micron sized) particles will jiggle around in a liquid. i.e. doing a random walk.**

This is predicted on theoretical grounds, but had neither been observed nor expected.

Now it is easy to observe!



Main points

1. “Visible” (ie, micron sized) particles will jiggle around in a liquid. i.e. doing a random walk.

This is predicted on theoretical grounds, but had neither been observed nor expected.

2. Calculation of diffusion constant of the particles

$$D = \frac{k_B T}{6\pi\eta a} = \frac{RT}{N 6\pi\eta a}$$

$$\lambda_x = \sqrt{2Dt}.$$

$$\lambda_x = \sqrt{t} \cdot \sqrt{\frac{RT}{N} \frac{1}{3\pi kP}}.$$

We now wish to calculate the magnitude of λ_x for one second if N is taken to be $6 \cdot 10^{23}$ in accordance with the results of the kinetic theory of gases; water at 17°C ($k = 1.35 \cdot 10^{-2}$) shall be chosen as the liquid, and the diameter of the particles shall be 0.001 mm. We obtain

$$\lambda_x = 8 \cdot 10^{-5} \text{ cm} = 0.8 \text{ micron}.$$

Thus, the mean displacement in 1 min. would be about 6 microns.

Random Walks in Stock Market Prices

by Eugene F. Fama

This is a review article (not posted) ...

Random walk theorists usually start from the premise that the major security exchanges are good examples of “efficient” markets. An “efficient” market is defined as a market where there are large numbers of rational, profit-maximizers actively competing, with each trying to predict future market values of individual securities, and where important current information is almost freely available to all participants.

In an efficient market, competition among the many intelligent participants leads to a situation where, at any point in time, actual prices of individual securities already reflect the effects of information based both on events that have already occurred and on events which, as of now, the market expects to take place in the future. In other words, in an efficient market at any point in time the actual price of a security will be a good estimate of its intrinsic value.

Now in an uncertain world the intrinsic value of a security can never be determined exactly. Thus there is always room for disagreement among market participants concerning just what the intrinsic value of an individual security is, and such disagreement will give rise to discrepancies between actual prices and intrinsic values. In an efficient market, however, the actions of the many competing participants should cause the actual price of a security to wander randomly about its intrinsic value. If the discrepancies between actual prices

It is unlikely that the random walk hypothesis provides an exact description of the behavior of stock market prices. For practical purposes, however, the model may be acceptable even though it does not fit the facts exactly. Thus although successive price changes may not be strictly independent, the actual amount of dependence may be so small as to be unimportant.

What should be classified as unimportant depends, of course, on the question at hand. For the stock market trader or investor the criterion is obvious: The independence assumption of the random walk model is valid as long as knowledge of the past behavior of the series of price changes cannot be used to increase expected gains. More specifically, if successive price changes for a given security are independent, there is no problem in timing purchases and sales of that security. A simple policy of buying and holding the security will be as good as any more complicated mechanical procedure for timing purchases and sales. This implies that, for investment purposes, the independence assumption of the random walk model is an adequate description of reality as long as the actual degree of dependence in series of price changes is not sufficient to make the expected profits of any more “sophisticated” mechanical trading rule or chartist technique greater than the expected profits under a naive buy-and-hold policy.